Junior Data Scientist

Data Science Team · Flock Energy

Experience	0–3 years
Employment type	Full-time
Working arrangement	Office-based
Location	Noida, India
Compensation	₹8–15 LPA, based on experience and capability

ABOUT FLOCK ENERGY

Flock Energy is a product-based engineering company building AI/ML solutions for electrical utilities. Incorporated in London in 2023 and operating in India ever since, we design innovative products – from in-house IoT hardware to large-scale data platforms – that solve real problems faced by an ever-evolving grid. We work with leading DISCOMs such as JVVNL, Tata Power and BSES, and partner with sector bodies including the Ministry of Power, REC and RRECL.

THE ROLE

We are looking for a **Junior Data Scientist** to join our growing data science team. Our work combines energy data, machine learning and optimisation – we analyse smart-meter and other time-series data to understand electricity consumption, identify patterns and develop practical energy insights.

You will work with experienced data scientists and engineers to analyse energy and sensor data, develop machine-learning models and turn experimental work into reliable product capabilities. This is a hands-on role suited to someone who enjoys solving real-world problems with imperfect data and wants to build strong foundations in applied data science.

WHAT YOU'LL WORK ON

- Explore, clean and analyse **smart-meter, energy and sensor datasets**.
- Work with **high-frequency and time-series data** to identify patterns, anomalies and changes in consumption.
- Engineer and evaluate features for **forecasting, classification, anomaly-detection and disaggregation** models.
- Train, test and compare statistical and machine-learning models.
- Support the development of **Non-Intrusive Load Monitoring (NILM)** models that estimate appliance or equipment-level consumption from aggregate electricity data.
- Build clear and reproducible data-processing and modelling pipelines in **Python**.
- Evaluate model performance using appropriate metrics and communicate the results clearly.

- Help investigate model failures, unusual data and differences between laboratory results and real-world performance.
- Work with software and data engineers to integrate models into production systems.
- Document experiments, assumptions, datasets and modelling decisions.
- Contribute to code reviews, team discussions and continuous improvement of our data science practices.
- Learn about electricity systems, demand flexibility and energy-market applications.

You do not need previous energy-sector experience. We are more interested in your analytical ability, Python skills, understanding of machine learning and willingness to learn.

WHAT WE'RE LOOKING FOR

- **0–3 years** of relevant professional experience. Strong graduates and candidates with relevant internships or substantial academic projects are welcome to apply.
- A degree in data science, computer science, statistics, mathematics, engineering, physics or another quantitative subject.
- Good working knowledge of **Python**.
- Experience using common data-science libraries such as **pandas, NumPy and scikit-learn**.
- Understanding of **core machine-learning concepts**, including training and test splits, feature engineering, model selection, overfitting and performance evaluation.
- Experience analysing structured datasets and presenting findings clearly.
- Familiarity with **SQL** and relational data.
- Good analytical and problem-solving skills, attention to detail and a willingness to investigate unexpected results.
- Ability to explain technical work clearly to both technical and non-technical colleagues.
- Ability to work collaboratively and receive constructive feedback.

GOOD TO HAVE

Any of the following would be useful, but none is required:

- Time-series analysis or forecasting; signal processing.
- Smart-meter, energy, **IoT**, telemetry or industrial sensor data.
- Anomaly detection, predictive maintenance or equipment monitoring.
- Non-Intrusive Load Monitoring or energy disaggregation.
- **LightGBM, XGBoost, PyTorch** or **TensorFlow**.
- Data visualisation using tools such as Matplotlib, Seaborn or Plotly.
- Cloud platforms such as **AWS, Azure** or Google Cloud.
- Git, Docker, APIs or basic MLOps practices.
- Experience deploying or monitoring a model in production.

- Knowledge of electricity consumption, energy systems or demand flexibility.

WHAT SUCCESS LOOKS LIKE

During your first year, you will:

- Develop a strong understanding of Flock Energy's datasets, products and modelling challenges.
- Deliver reliable exploratory analyses and clearly documented experiments.
- Contribute useful features and improvements to our modelling pipelines.
- Build and evaluate models with appropriate guidance from senior team members.
- Progressively take **ownership** of defined data-science problems.
- Communicate your findings clearly and contribute positively to the wider product and engineering team.

WHAT WE OFFER

- A salary of **₹8–15 LPA**, based on experience and demonstrated capability.
- The opportunity to work on practical machine-learning problems in the energy sector.
- Regular support and mentorship from experienced data scientists and engineers.
- Exposure to the **full modelling lifecycle**, from raw data and experimentation to deployment and monitoring.
- Meaningful responsibility and opportunities to grow as the company and team expand.
- A collaborative environment where careful analysis, curiosity and continuous learning are valued.

HOW TO APPLY

Make your resume along with a short explanation of a data-science, machine-learning or analytical project you worked on. Where possible, explain the problem, your personal contribution, the methods you used and what you learned from the results.

Flock Energy welcomes applicants from different backgrounds. If you meet most – but not necessarily all – of the requirements and are excited by the role, we encourage you to apply.